



Investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine - NTSB-style probable cause report

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INVESTIGATION WORKFLOW SUITE

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Executive summary

What happened

At 02:47 local time on 11/08/2026, an empty 220 t class rigid haul truck descending the main ramp of the Stage 4 open pit at Bindarra Gold Mine struck the right side of a light vehicle emerging from the RL 312 dewatering spur, at 24 km/h. The truck pushed the light vehicle 18.4 m down the ramp and rolled it onto its left side against the inner windrow.

The driver, a survey subcontractor's grade control technician on his third night shift at the mine, sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. He remained conscious, was extricated through the rear 27 minutes after the collision and was flown to the metropolitan trauma centre at 07:40. His seatbelt was fastened; the roll-over protection structure deformed without being breached.

The light vehicle was destroyed. The truck, damaged at the front left, returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes and the ramp below RL 340 was closed for 22 hours. The occurrence was notified to the State Mines Inspectorate as a reportable injury.

What the investigation found

The probable cause was very likely a new intersection opened into an operating haul road without the controls that should have followed it, so both vehicles arrived there with neither operator aware of the other.

Four factors very likely contributed:

- **No management of change.** The spur's works order recorded management of change as not required, so nothing was referred to the traffic management plan owner, the in-cab road network or the geofence configuration.
- **Inadequate sight line.** The spur met the ramp 34 m beyond a crest, giving 41 m of sight line against a plan minimum of 120 m and a stopping distance of 96 m.
- **No positive call required.** Positive radio calls were required only at intersections listed at the annual review.
- **Entry without stopping.** The light vehicle entered without stopping or establishing the truck's position. The mouth had no sign, stop line or give-way control; a full stop would have added only a few metres of sight line.

Four further factors likely contributed:

- **Suppressed alerts.** Collision awareness alerts were suppressed inside a geofence extended in June 2026 over the intersection. The alert would have come after braking began; its influence lay in the operator's expectation, built by months of silence on that ramp, that silence meant an empty road.
- **No radio call made.** Neither operator made a positive call, consistent with the plan and local practice.
- **Unverified induction.** The contractor induction was accepted as equivalent; the site supplement and pit driving assessment were never verified.
- **Discretion in the guideline.** MSG-14 left the review trigger undefined and interaction zones to the site. The mine was not in breach, and this finding is less firm than the others.

The truck operator braked and steered into the windrow within 1.4 seconds of sighting, lowering impact speed from 37 km/h to 24 km/h and keeping the light vehicle against the wall side.

A wet ramp and degraded brakes were ruled out. Whether dust reduced visibility could not be determined: particulate was elevated, but the light vehicle driver described clear air, the sighting distance matched clear-air geometry, and nothing at the site measures visibility.

Two factors did not contribute:

- **Night rescue roster.** One paramedic was rostered at night, the second responder coming from the village, delaying extrication by approximately 9 minutes without affecting the clinical outcome. It very likely increases risk.
- **Operator fatigue exposure.** The operator was on night 10 of 14, in the circadian low, which likely increases risk. His recorded response showed no degradation.

What can be done

Six safety issues were identified:

- a management of change exclusion that let a new intersection enter an operating pit without referral
- a positive call list that could change only once a year
- a suppression geofence moved without reference to the traffic management plan
- an access card system unable to verify pit driving competence
- guidance leaving key terms for each site to define
- a night extrication capability that does not scale with the casualty

A new road changes traffic risk the day it is cut; controls tied to an annual review, or to a classification that removes referral, will not follow it. No safety action had been recorded against any of these issues.

Factual information

History of the occurrence

Times are local (UTC plus eight hours), except under Recorded data, where they are UTC as logged.

The RL 312 dewatering spur was a 240 m access road, 8 m wide with windrows on both sides, to dewatering bore DW-14 in the Stage 4 open pit at Bindarra Gold Mine. Works under WO-26-4471 were completed on 31/07/2026. On 02/08/2026 the bore was energised, light vehicles first used the spur, and the works log recorded removal of the "ROAD CLOSED - CONSTRUCTION" sign at the spur mouth. Nothing replaced it.

At about 22:10 on 04/08/2026, a dewatering technician drove out of the spur in front of a descending haul truck he estimated at about 60 m away, reversed and let it pass. That night he submitted hazard report HAZ-2026-0917, describing the rise in the ramp, the windrow and the absence of any sign or give-way control, and wrote: "I do not think they know the road is there."

The Dewatering Superintendent closed the report on 06/08/2026, recording that the crew had been reminded at pre-start on 05/08 to take care and give way to haul trucks: "No further action required. Closed." The risk rating was reduced from Medium to Low. The report was not referred to the mining department, the contractor, the survey department or the traffic management plan owner.

On the night of the occurrence, Stage 4 crews returned from crib at 02:22. Haul truck HT-407, a 220 t class rigid haul truck operated by Warrigal Mining Services, was on its second post-crib descent of the main ramp, empty.

Light vehicle LV-118 was driven by a grade control technician employed by Merrivale Survey & Geotechnical, its only occupant. He had completed levels at the DW-14 bore collar and drove out along the spur. [NOT ESTABLISHED: time LV-118 left the bore collar] No permit, call-in or board existed by which the shift would have known he was in the pit, and the night shift supervisor did not know.

The driver said he slowed at the spur mouth, looked left up the ramp, saw nothing and drove out. He believed he did not stop completely and did not recall hearing anything on the radio. The truck operator reported first seeing the light vehicle's beacon whip at about 40 m as he came over the crest above RL 312, and braked and steered right towards the windrow. Neither driver made a positive radio call before entering the intersection.

At 02:47 the truck's front left struck the right side of LV-118 at 24 km/h, pushing it 18.4 m down the ramp and rolling it onto its left side against the inner windrow. The truck stopped below the intersection.

The operator called dispatch 29 seconds after the collision. Dispatch stopped all Stage 4 units at 02:48:01 and held the operator in his cab until releasing him at 02:48:44. The emergency response team was paged at 02:48. The first response vehicle entered the pit at 02:51 and reached the scene at 02:54. The second responder arrived at 03:07 with the hydraulic rescue tool.

The driver, conscious throughout, was extricated through the rear of the vehicle at 03:14, 27 minutes after the collision, and reached the surface medical facility at 03:41. He had a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was flown to the metropolitan trauma centre at 07:40.

Immediate actions:

- The Stage 4 pit was stood down for 9 hours 15 minutes and the main ramp below RL 340 closed for 22 hours.

- The spur was barricaded and both vehicles left in position.
- The truck operator and shift supervisor were stood down and tested for drugs and alcohol; both results were negative.
- At 05:10 the Mining Manager directed a positive call on the pit channel before entering any Stage 4 intersection, whether listed in the plan or not.

The occurrence was notified to the State Mines Inspectorate as a reportable injury.

Personnel information

Haul truck operator. 11 years in open pit mining, 7 on 220 t class rigid haul trucks; 14 months at Bindarra and 4 years 2 months with Warrigal Mining Services. Verification of competency on type issued 06/2025; annual refresher 02/2026. His site general induction (06/2025) included a 23 slide traffic management module naming the nine Appendix B intersections and not addressing unlisted intersections. No later traffic management refresher was recorded.

The occurrence was on night 10 of a 14 night swing, 10 hours 47 minutes into a 12 hour 30 minute shift, after 6 hours 30 minutes of sleep. The operator described himself as tired in the way usual for night 10, and awake and driving properly.

Light vehicle driver. 6 years in grade control, 4 with Merrivale. He completed Merrivale's two hour online general mining induction on 24/07/2026. His Bindarra site access card was issued on 09/08/2026 under Merrivale's recognition agreement, after a 20 minute gatehouse briefing on muster points, speed limits and personal protective equipment. There is no record that he received the Bindarra site-specific supplement or undertook the Bindarra light vehicle pit driving assessment; the gatehouse required neither. He had not heard of the assessment, had never seen the traffic management plan, and nobody had told him to call before entering the ramp.

It was his third night shift at Bindarra, on a 9 on 5 off roster, 18:00 to 06:00. No duty or rest concerns were identified. He had driven the spur six or seven times in the preceding week without meeting a haul truck.

Night shift supervisor. 19 years in open pit mining, 2 years 6 months in the role, on night 10 of 14. He was responsible for the haul, ancillary and dozer fleets and dispatch, not for the dewatering crew, the survey contractor or other pit light vehicles.

No site training record, for any person, covered the collision awareness system's suppression geofences.

Aircraft and equipment information

HT-407. Mean deceleration from brake application to impact was 1.85 m/s², against 1.9 m/s² in the monthly brake test on 03/08/2026. Retarder and service brakes performed as specified. Damage was confined to the front left bumper, left mudguard and lower two access ladder steps. The truck returned to service on 14/08/2026.

The collision awareness system operated at Level 7 (operator advisory); Level 9 (automated intervention) was available on the fleet but not enabled at Bindarra. Advisory alerts were raised when a tag entered the 60 m detection ring closing above 10 km/h, except inside a suppression geofence.

The S4-LOAD geofence was drawn on 14/03/2025 around the Stage 4 loading face to stop nuisance alerts. On 09/06/2026 the contractor's technical services group extended it northward as a configuration task, without review against the traffic management plan. Its boundary lay 180 m up-ramp of the RL 312 intersection. The standing alert statistics report excluded suppressed events.

The fleet management system in-cab map for Stage 4, last published on 24/07/2026, did not show the spur or any intersection at RL 312.

LV-118. No fleet management system; a passive collision awareness tag. The vehicle was destroyed. The roll-over protection structure was deformed inward at the right rear mounting but not breached. The driver's seatbelt was fastened, with loading marks on the webbing. The 4.5 m amber beacon whip was sheared at its base; its lamp worked when tested.

Meteorological information

[NOT ESTABLISHED: recorded meteorological observations]

It was night and the ramp was dry. The truck operator said some dust was hanging and he could not say whether it made a difference. The light vehicle driver said the air was clear, with no dust in the spur, and he could see the ramp surface and windrow.

Recorded data

Fleet management system download, HT-407 (UTC):

Time	Speed	Position / event
18:45:58	41 km/h	RL 368
18:46:31	39 km/h	RL 348, retarder on second position
18:47:06	37 km/h	RL 314, 41 m above the intersection; last steady data point
18:47:07.4	37 km/h	Brake pressure 0 to 620 kPa in 0.4 s; retarder maximum; steering 14 degrees right
18:47:09.2	32 km/h	Collision awareness proximity event
18:47:12.0	24 km/h	Collision
18:47:14.3	0 km/h	19 m below the intersection; park brake applied

The download showed a response time of 1.4 seconds. Without braking, collision speed would have been 37 km/h.

The collision awareness log recorded tag LV-118 at 18:47:09.200 at 21.4 m, closing at 13 km/h, in zone S4-LOAD: "alert SUPPRESSED (geofence)", followed by an impact of 1.9 g at 18:47:12.000. Outside the geofence, the event would have raised an advisory tone and screen indication.

The pit channel recording (RAD-2026-0332) captured the operator's first call at 18:47:41 and the stop-all call at 18:48:01.

Aerodrome and site information

At RL 312 the main ramp ran approximately north-west on a 9.1 per cent descending grade, 24 m wide and well maintained. The spur joined from the east, on the inside of a left-hand bend, at 71 degrees, 34 m beyond the high point of a crest.

Surveyed sight lines:

- A truck operator at 5.80 m eye height first saw the spur mouth at 41 m.
- A driver at the spur mouth at 1.45 m eye height first saw a descending truck at 38 m.
- The 1.6 m inner windrow obscured the truck's lower body for the last 25 m of the spur.

Calculated stopping distance of an empty 220 t class truck at the posted 40 km/h on a 9 per cent descent was 96 m.

The intersection had no warning sign, give-way or stop control, delineation or posted spur speed.

HT-407's braked left front scuff began 38.5 m up-ramp of the spur mouth and ran 62 m. The impact point was 6.2 m into the ramp and 3.1 m from the inner windrow toe. The LV-118 gouge ran 18.4 m, changing to a body panel gouge at 11.2 m.

Organisational and management information

Corella Resources' traffic management plan, BIN-TMP-004:

- Clause 6.2 required intersection sight distance not less than the largest vehicle's stopping distance, and never less than 120 m.
- Clause 6.4 required advance warning signs, give-way control and delineation, but only at intersections on the plan's road layout drawing, which did not include the spur.
- Positive radio calls were required at the nine Appendix B intersections, which did not include the spur.

The truck operator and supervisor each described calling at those nine intersections and nowhere else as pit practice.

The plan was reviewed annually on 14/11/2023, 10/11/2024 and 12/11/2025. No out-of-cycle review under clause 9.1 had occurred in its eleven years. Since the 2025 review, 14 new roads had been cut in Stage 4. The 3 cut under a management of change record were referred to the plan owner; the 11 cut under works orders, including the spur, were not.

WO-26-4471 classified the work as routine earthworks: "Management of change not required, CRL-PRO-022 clause 5.3". It had no scope for signage, survey pickup, plan referral, road network publication or geofence review, and its handover field read "N/A". Publishing a revised in-cab road network was a survey task raised by a management of change record.

There was no process for a light vehicle entering an operating pit.

The night shift was rostered with one mines rescue paramedic. The second responder, required before any extrication, was called from the accommodation village 11 km from the gatehouse. The post-incident review found extrication was delayed by approximately 9 minutes and that the delay did not affect the clinical outcome on this occasion.

Additional information

A search of Corella Resources hazard reports raised at Bindarra in the 24 months to 11/08/2026 returned four relevant reports, including HAZ-2026-0917. The others, from 03/2025, 09/2025 and 02/2026, each described a light vehicle and haul truck meeting unexpectedly at a different intersection not listed at Appendix B. All four were closed within three days by reminding crews to take care and noting that heavy vehicles have right of way. None was referred to the plan owner or changed controls.

Analysis

External Influences

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

The mine's traffic management plan was made under Mines Safety Guideline MSG-14, revision 4. The guideline is one way of meeting regulation 6.22 of the Mining Operations Safety Regulations 2018. That regulation requires only that "the operator shall manage the risks arising from the interaction of vehicles at the mine". It was virtually certain that the guideline left three terms for the site to define:

- Clause 3.2 requires review "following any significant change to the mine's road network" but does not define a significant change or give an example of one.
- Clause 5.1 requires the operator to nominate the interaction zones where positive communication is needed. It sets no minimum and no basis for the nomination.
- Clause 5.4 lists five means of making a light vehicle detectable to a heavy vehicle operator and does not say which, or how many, must be provided.

Under clause 3.5, the plan "is not approved by, and need not be submitted to, the Inspectorate" except on request after an occurrence or inspection.

Bindarra's plan met the guideline on its face. It had been reviewed within twelve months. It nominated nine interaction zones. It provided for detectability by line of sight and elevated marking. The mine was not in breach. The matter analysed here is the width of the discretion the guideline leaves. It is not a failure by the Inspectorate, and it is not a failure by the mine to comply.

At Bindarra, all three discretions were filled by BIN-TMP-004, which is reviewed once a year and has no outside review. Each one led back to the annual cycle:

- Clause 9.1 repeated the guideline's undefined review trigger.
- An intersection could reach Appendix B only through the road layout drawing at Schedule 1, which was reissued only at the annual review.
- No provision allowed an intersection to be added to either between reviews.
- No provision required a new road to be referred to the plan's owner.

Detectability rested partly on line of sight. At the RL 312 intersection the line of sight was 41 m, against the plan's own minimum of 120 m. That geometry is analysed under Local Conditions.

The document register and the reporting system show how this worked in practice. Fourteen new roads were cut in the Stage 4 pit after the 12/11/2025 review. The three cut under a management of change record were referred to the plan's owner. None of the eleven cut under a works order were. No out-of-cycle review had been held in the plan's eleven years. In the 24 months before the occurrence, four hazard reports described a light vehicle and a haul truck meeting unexpectedly at an unlisted intersection, at four different locations. One of these was HAZ-2026-0917, which concerned the RL 312 intersection. Each was closed within three days with a reminder to the crew. None was referred to the plan's owner, and none led to a change in Appendix B, Schedule 1 or any physical control.

The case for influence was as follows. A defined review trigger and a stated minimum for interaction zones would have given the eleven unrefereed roads somewhere to go. Each internal control that failed in this occurrence failed for want of a mechanism that brought a new road to the plan's owner. A clearer external requirement would have been a reason to build that mechanism.

The case against influence carried real weight. The guideline did not prevent anything. Bindarra could have held an out-of-cycle review under its own clause 9.1 at any time. MSG-14 did not stop the mine from listing every intersection it had. On this reading, the mine did not fail to act because the guideline was vague. It failed to act because nothing inside the mine put the question to the plan's owner, and the guideline is only background. The mine's handling of the spur under a works order is analysed under Organisational Influences. The limit of positive calls to listed intersections is analysed under Risk Controls.

These two readings were weighed. The investigation accepted that the guideline's discretion likely contributed, on the balance of the argument. This conclusion rests on a narrower margin than the other contributing factors in this occurrence. The contrary reading was not disproved. The deciding point was the pattern: across eleven roads, four hazard reports and eleven years, the same undefined terms were filled the same way, with no outside check. This is more consistent with a discretion that set no floor than with one isolated lapse at one site.

The breadth of the traffic management guidance, which left the review trigger undefined and the interaction zones and means of detectability to the site, likely contributed to the RL 312 spur being cut and opened without referral to the plan's owner and to positive calls being required only at intersections listed once a year.

Organisational Influences

Access spur cut and opened to traffic under a works order, with no management of change raised

CRL-PRO-022 clause 5.2 requires a management of change record before "a change to the mine's road network". Clause 5.3 removes that requirement for "routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced."

WO-26-4471 cut 240 m of access road to bore DW-14 at RL 312. It classified the work as "Routine earthworks within the approved Stage 4 design" and recorded "Management of change not required, CRL-PRO-022 clause 5.3." The procedure, the works order and the survey department's road register agree with each other. It was virtually certain that the spur was built and opened under a works order with no management of change raised.

The consequence lies in clause 5.4. A management of change record requires three referrals:

- to the traffic management plan owner;
- to the survey department, for the fleet management system road network;
- to the contractor's technical services group, for the collision awareness system configuration.

The works order form at Schedule 4 has no equivalent fields. Its handover field was the only place any follow-up could have been recorded, and it reads "N/A". None of the three referrals was made.

Each system the referrals serve was found unchanged on the night:

- The in-cab road network was last published on 24/07/2026, seven days before the spur was cut. It showed the main ramp with no intersection at RL 312.
- The spur was not on the traffic management plan's road layout drawing.
- The collision awareness system geofences had not been reviewed against the new intersection.

This pattern was not confined to one road. Fourteen new roads were cut in Stage 4 after the last plan review. The three raised under a management of change record were referred to the plan's owner. None of the eleven raised under a works order was.

One argument runs against this factor's influence. A referral is not a control. Had a management of change record been raised, the plan's owner might have considered the spur and still left it off the list. The investigation weighed this against the fact that the record was the only route by which the intersection could have reached the plan, the in-cab map or the geofence configuration. Its absence left all three untouched. The possibility of an inadequate response to a referral reduces the confidence in this factor's influence but does not remove it.

The consequences of the unlisted intersection are argued elsewhere in this analysis: the 41 m sight line, the absence of any positive-call requirement, and the suppressed collision alert. This subsection establishes only that each of them traces back to the works order classification.

It was very likely that opening the spur under a works order, without a management of change record, contributed to the collision by leaving the new intersection outside the traffic management plan, the in-cab road network and the collision awareness system review.

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

CRL-STD-009 clause 7.3 allows a contractor's own general induction to be accepted as equivalent where the contractor holds a recognition agreement. Merrivale Survey & Geotechnical had held one since 2023.

Clause 7.4 sets the limit of that equivalence. It "does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment". Both "shall be completed at site by every person who will drive a vehicle in an operating pit and shall not be substituted." Clause 7.5 makes the engaging department responsible for verifying this before a site access card is issued.

The standard therefore required the right things. The failure lay in where verification could happen:

- The gatehouse issued cards against the recognition agreement alone.
- It held no record of the supplement or the assessment, and was not asked for one.
- Nothing in the access card system distinguished a person permitted to drive in an operating pit from one who was not.

The light vehicle driver's training file showed:

- the Merrivale online induction module, completed on 24/07/2026;
- a 20 minute gatehouse briefing on muster points, speed limits and personal protective equipment;
- no site-specific supplement and no pit driving assessment.

He had never seen BIN-TMP-004 and did not know it existed. The occurrence was on his third night shift at the mine.

This evidence comes from the standard, the site training files and the gatehouse card register. Merrivale's own records for the site supplement were not examined, and the driver was not asked about it. It was therefore extremely likely, rather than certain, that he drove in the operating pit without either item having been completed or verified.

The pit driving assessment is where a light vehicle driver learns how the pit's traffic rules work in practice. This includes give-way, positive calls and what to do at a blind intersection. That is the knowledge bearing on how he entered the ramp from the spur, which is argued under the individual actions.

Two considerations limit this factor's influence:

- The central rule the assessment would have taught, that light vehicles give way at all times, is one the driver was already applying.

- The intersection's 41 m sight line would have defeated attention on its own, whatever the driver had been taught.

For these reasons the influence was assessed as likely rather than higher.

It was likely that the absence of any point at which the pit driving assessment and site supplement were verified contributed to the collision, by leaving the driver without the site's instruction on entering a haul road at a blind intersection.

Risk Controls

Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

The collision awareness system on HT-407 detected LV-118 and presented nothing in the cab. The log recorded a proximity event at 18:47:09.2, at a range of 21.4 m and closing at 13 km/h, marked "alert SUPPRESSED (geofence)". Outside the S4-LOAD geofence, the same event would have produced an advisory tone and a screen indication. The configuration, the event log and the fleet statistics came from the system itself and were downloaded sealed. It is virtually certain that the intersection lay inside a suppressed area and that no alert was given.

The geofence itself was not the deficiency. It had been drawn in 03/2025 because nuisance alerts around the loading face had made the system unusable. The deficiency was the extension on 09/06/2026, which moved the boundary to 180 m up-ramp of the RL 312 intersection. The contractor's technical services group made that change as a configuration task. The change was not recorded in the traffic management plan or reviewed against it, and no one on site had been trained on the geofences.

The evidence against influence was strong. An alert would have sounded 1.8 seconds after the operator began braking and 2.8 seconds before impact, so it would have told him nothing he did not already know. The system was advisory only, because Level 9 intervention is not enabled at Bindarra. With 41 m of sight line, the outcome was fixed before either the sighting or the alert. On that reading, the suppression was a serious deficiency in the fleet but not a cause of this collision.

The investigation accepted a different mechanism: the expectation that the silence had created. In the 12 months before the occurrence, HT-407 raised 412 advisory alerts, all of them outside the geofence. Across the fleet, 68 proximity events inside the geofence were suppressed. Nine of them involved a light vehicle tag on the main ramp below the boundary, which was the configuration of this occurrence. The standing report covered alert counts and system health but not suppressions, so none of these events was reported to anyone. The operator and the supervisor each raised the matter without being asked. In his own account, the operator described silence as having come to mean an empty road. That expectation was a condition he carried towards the crest. Both readings are recorded here. The accepted mechanism rests on the operator's expectation, not on the timing of the alert.

It was likely that suppressing alerts across the RL 312 intersection, through a geofence extension made without reference to the traffic management plan, contributed to neither operator being aware of the other, by leading the truck operator to expect a clear ramp rather than through the absent final alert.

Positive radio calls required only at listed intersections, and the list is rewritten once a year

BIN-TMP-004 clause 4.3.1 requires a positive call before entering any intersection listed at Appendix B. Entry is permitted only when the call "is acknowledged or the channel is confirmed

clear". Clause 4.3.2 requires no call at any other intersection. The RL 312 spur appeared neither at Appendix B nor on the Schedule 1 drawing. At this intersection, therefore, the mine's own defence against this interaction was inactive by its own terms. This is documentary and virtually certain.

The trigger for adding an intersection is circular:

- an intersection reaches Appendix B only through Schedule 1;
- Schedule 1 is reissued only at the annual review;
- no provision allows either document to change between reviews; and
- no provision requires a new road to be referred to the plan's owner.

Clause 9.1 does provide for a review after significant change, but no out-of-cycle review has been held in the plan's eleven-year history. Of the 14 roads cut in Stage 4 since the 12/11/2025 review, the 11 cut under a works order were never referred. The earliest date on which the spur could have been listed was 12/11/2026, three months after the collision. How the spur came to be opened without referral is argued under Organisational Influences. The guidance that leaves the trigger to the site is argued under External Influences.

Practice matched the plan's wording. The truck operator, the light vehicle driver and the night shift supervisor each independently described calls being made at the nine intersections on the crib room board and nowhere else. The truck operator's induction named those nine intersections and said nothing about any others. The pit channel recording contains no call from either vehicle about the spur, the ramp or a light vehicle in the pit. That neither operator made a call is a separate finding.

The reporting system had already exposed the gap. In the 24 months before the occurrence, three hazard reports, from 03/2025, 09/2025 and 02/2026, each described a light vehicle and a haul truck meeting unexpectedly at an unlisted intersection. HAZ-2026-0917 described a near miss at this intersection seven days before the collision. All four reports were closed within three days with a reminder that heavy vehicles have right of way. None was referred to the plan's owner, and none led to a change to Appendix B, Schedule 1 or any physical control.

Had a call been required, it would very likely have been effective. The channel was open and both vehicles were on it. At 18:46:18, dispatch was tracking the truck closely enough to place it about four minutes from the loading face. The remaining doubt is that a call made but not acknowledged would still have left the driver to decide whether to proceed.

It was very likely that the plan's restriction of positive calls to a list that could change only at the annual review contributed to neither operator making a call at the RL 312 intersection.

One emergency response paramedic rostered on night shift, with the second responder called from the village

The night shift is rostered with one mines rescue paramedic. A second responder must be present before extrication begins. That responder is called in from the accommodation village, 11 km from the gatehouse, and brings the hydraulic rescue tool. The post-incident review found that extrication was delayed by approximately 9 minutes while waiting for the second responder and the tool. The roster, the response log and the review agree, and the arrangement virtually certainly existed.

The arrangement had no bearing on this occurrence or its outcome. The collision and the injuries were complete before the team was paged. The review recorded, on clinical grounds, that the delay did not affect the outcome. The driver was conscious and stable, his seatbelt was fastened, and the roll-over protection structure had not been breached.

The significance of the arrangement is structural rather than clinical. A fuel fire, an unconscious casualty, or an airway needing access within the first ten minutes would have turned the same nine

minutes into a different outcome. Nothing in the arrangement scales with the severity of the event, and the gap exists on every night shift.

It was very likely that rostering a single paramedic on night shift, with the second responder and the rescue tool called in from the village, increased the risk of delayed extrication of a seriously injured person.

Local Conditions

The spur meets the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m

The mine survey department picked up the spur mouth 34 m beyond the high point of a crest in the ramp. From a descending truck at an eye height of 5.80 m, the first line of sight to the spur mouth was at 41 m. From the spur mouth at 1.45 m, the first line of sight to a descending truck was at 38 m. For the last 25 m of the spur's approach, the 1.6 m inner windrow hid the truck's lower body.

Clause 6.2 of BIN-TMP-004 sets two limits. Sight distance must be not less than the stopping distance of the largest vehicle at the posted speed, and "in no case less than 120 m". The stopping distance of an empty 220 t class truck at 40 km/h on a 9 per cent descending grade was 96 m. The survey concluded: "The intersection does not meet clause 6.2 on either limb."

The sight line was less than half the stopping distance. At 37 km/h, 41 m gave the truck operator about four seconds. On the plan's own figures, no response the operator could have made would have stopped the truck within that distance. This limit came before attention, training or radio discipline. Those defences could only work at an intersection whose geometry already allowed a stop, and this one did not.

Nothing on the ground made up for the geometry. The intersection had no warning sign, no give-way control, no delineation and no posted speed. Clause 6.4 required each of these, but only at intersections shown on the plan's road layout drawing, and the spur was not on it. How the spur came to be opened outside the plan is addressed under Organisational Influences.

The hazard was known in advance. Seven days before the collision, a dewatering technician described the same sight line from the same position: "you cannot see up the ramp from that road". The hazard report and the survey were independent sources, and they agreed.

It is virtually certain that the spur met the ramp with 41 m of sight line against a stopping distance of 96 m and a plan minimum of 120 m, and very likely that this contributed to both vehicles reaching the intersection with neither operator aware of the other.

Dust on the ramp reduced the truck operator's ability to see the light vehicle

Dust was examined as a possible reason the light vehicle was not seen sooner. The question could not be resolved.

Some evidence suggested a deviation from normal conditions:

- The last water cart pass on that section was 1 hour 37 minutes before the collision. The standing interval was 90 minutes, so a pass was 7 minutes overdue.
- The fixed dust monitor at the ramp crest, 180 m up-ramp of the intersection, recorded PM10 of 142 micrograms per cubic metre in the ten minutes covering the collision. The shift mean was 88.
- The truck operator described dust hanging on the ramp.

- He was descending behind his own lights into an unlit pit, the condition in which airborne dust most affects a driver.

Other evidence did not support an effect on what could be seen:

- The light vehicle driver was firm that the air was clear and that he could see the ramp surface and the windrow.
- The truck operator declined to say the dust had made a difference.
- The monitor reading was well below the site trigger of 250. It measured particulate at a monitor head 180 m away, not visibility from either cab.
- Nothing at Bindarra measures visibility.

The strongest point against an effect was the sighting distance itself. The truck operator saw the light vehicle at 41 m, which is what the survey gives for perfectly clear air. The account and the geometry therefore agree without dust having to explain anything.

The deviation was real and was in the right place at the right time. Its effect on detection could not be established either way. Settling it would need a measurement of visibility from the cab on the night, and none exists. Whether dust reduced the truck operator's ability to see the light vehicle remains unresolved, and nothing is drawn from it.

The ramp surface was wet and the truck could not stop in the distance available

A wet surface was ruled out as very unlikely, on three independent sources:

- the met station recorded no precipitation, and the last rainfall at the site was 26 days before;
- the investigator found the surface dry throughout at 06:20 local; and
- the truck achieved a mean deceleration of 1.85 m/s^2 , against its own tested dry figure of 1.9 m/s^2 .

Night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low

It is virtually certain that the truck operator was on night 10 of a 14-night swing, 10 hours 47 minutes into a 12 hour 30 minute shift, at 02:47 local, in the window of lowest circadian alertness. Every limit in the contractor's fatigue management standard was met. His own sleep record showed 6 hours 30 minutes of sleep in the preceding day. In interview he described himself as tired in the ordinary way for a tenth night but awake and driving properly.

Those are the conditions in which slowed detection and response would be expected, and the task was detection in an unlit pit. However, the fleet management system download showed a response of 1.4 seconds, inside the 1.5 seconds assumed in the plan's stopping distance calculation. The braking and steering that followed were immediate and in the only available direction. On the plan's stopping distance, the truck could not have stopped within 41 m however quickly the operator responded. There was no delayed or missed action through which fatigue could have acted. It is unlikely that fatigue influenced this occurrence.

It remains likely to matter. The roster places every haul truck operator on this pit, and on three other pits on the site, at the wheel of a 220 t vehicle in the small hours of a tenth consecutive night. At a mine with no fixed lighting in the pit, the operator's eyes are the only defence against an unlit hazard at an unlisted intersection.

Individual Actions

Neither operator made a positive call on the pit channel before entering the intersection

The pit channel was recorded continuously from the return from crib at 18:22:41 until after the collision. Neither vehicle transmitted anything about the RL 312 spur, the dewatering bore, or a light vehicle in the Stage 4 pit. Apart from logging on at 17:04, LV-118 made no transmission that shift. All three interviews agreed. It was virtually certain that no call was made.

No call was required. BIN-TMP-004 clause 4.3.2 states that at intersections not listed at Appendix B "No call is required". The RL 312 intersection was not listed. The truck operator, the light vehicle driver and the night shift supervisor each described the same practice independently. Calls were made at the nine intersections on the crib room board and nowhere else. The supervisor had never corrected anyone for not calling elsewhere. He considered that calls at every intersection would make the channel unusable. Local practice matched the plan exactly, so the omission belonged to no individual.

A call would have cost nothing. It would have reached the other vehicle, dispatch and the supervisor. Dispatch was already tracking the truck closely enough to place it: at 18:46:18 it told the excavator "407's about four minutes off you." A call would have given each operator knowledge of the other. It would not in itself have given them separation. The two vehicles would still have had to settle who went first over 41 m of sight line. For that reason the contribution was assessed as likely and no higher.

The rule that confined calls to listed intersections, and the annual rewriting of that list, are argued under Risk Controls and Organisational Influences.

It was likely that the absence of a positive call, which the plan did not require at this intersection, contributed to both vehicles arriving there with neither operator aware of the other.

Light vehicle entered the ramp from the spur without stopping and without establishing the truck's position

The driver's account was that he left the spur at about 25 km/h, looked left up the ramp, saw nothing and drove out. On read-back he corrected his account from stopped to slowed, which was against his own interest. He did not believe he had come to a complete stop. The scene evidence agreed with him: the impact was 6.2 m into the ramp from the eastern edge of the spur mouth. It was virtually certain that the light vehicle entered the ramp without stopping and without establishing the truck's position.

The driver knew the rule that applied. Clause 4.1 requires light vehicles to give way to heavy vehicles, and he described himself as applying it. The intersection gave him nothing to apply it with. There was no give-way control, stop line, warning sign or posted speed at the mouth, because clause 6.4 applies only to intersections on the road layout drawing. He had not completed the Bindarra pit driving assessment. He had never seen the traffic management plan. It was his third night shift at Bindarra.

Stopping would have changed little of what he could see. From the mouth, at his eye height of 1.45 m, his first line of sight to a descending truck was at 38 m. For the last 25 m of the spur's approach, the windrow hid the truck's lower body. A look left from a stop would have shown him approximately what his look at 25 km/h showed him. The truck needed 96 m to stop.

The entry was the proximate step, because with no light vehicle on the ramp there would have been no collision. The contribution was assessed as very likely rather than higher because a full stop would have gained only a few metres of sight line. The induction arrangement under which his site supplement and pit driving assessment were never verified is argued under Organisational Influences.

It was very likely that the light vehicle's entry to the ramp without stopping and without establishing the truck's position contributed to both vehicles arriving at the intersection with neither operator aware of the other. The intersection had no control and a 38 m sight line from the spur.

Truck operator braked and steered into the windrow within 1.4 seconds of sighting

The download showed the last steady data point at 18:47:06, at 37 km/h and 41 m above the intersection. The survey puts the truck operator's first line of sight to the spur mouth at the same 41 m. At 18:47:07.4 service brake pressure rose from 0 to 620 kPa in 0.4 seconds. In the same movement the retarder went to maximum and 14 degrees of right steering was applied. The scene scuff deviated 1.4 m right over its first 20 m, consistent with that input. It was virtually certain that the operator responded in 1.4 seconds. That was inside the 1.5 seconds allowed in the plan's own stopping distance calculation.

Mean deceleration was 1.85 m/s², against the unit's tested 1.9 m/s². Impact speed was 24 km/h. Without braking it would have been 37 km/h. The steering took the collision to the wall side of the road, the only direction that did not carry the light vehicle towards a 100 m drop. The light vehicle came to rest against the inner windrow.

The truck stopped 19 m below the intersection. The operator reported the collision and stayed in the cab on the dispatcher's instruction until the ramp was stopped. He then went to the driver and stayed with him until the response team arrived.

It was very likely that the truck operator's braking and steering, begun 1.4 seconds after first sighting, reduced the severity of the light vehicle driver's injuries.

Occurrence Events

Both vehicles arrived at the intersection with neither operator aware of the other

The contributing conditions examined in the preceding subsections came together at this point. They are best read as one mechanism, not as separate paths to the collision.

HT-407 descended the ramp empty at 37 km/h. It had four possible sources of warning, and none of them worked:

- The in-cab road network, last published on 24/07/2026, showed no intersection at RL 312.
- There was no warning sign on the approach.
- There was no call on the pit channel.
- The collision awareness system logged LV-118 and suppressed the alert. The log reads "range 21.4 m closing 13 km/h zone S4-LOAD alert SUPPRESSED (geofence)".

LV-118 left the spur at about 25 km/h. There was no give-way control at the mouth, no call, and the driver had no knowledge of the truck.

The night shift supervisor did not know a light vehicle was in the pit. No permit, call-in or board existed by which he could have known.

The survey placed the first possible sighting at 41 m for the truck operator and 38 m for the light vehicle driver. This is the geometry of a spur meeting the ramp 34 m beyond a crest, with a 1.6 m inner windrow along the last 25 m of the spur. The download recorded the truck's last steady data point 41 m above the intersection, which agrees with the survey figure.

Three independent recordings and all three interviews agreed: neither operator knew the other was there before the first sighting. One limitation applies. LV-118 carried no fleet management system, and its CAS transponder was a passive tag. Its movements before the collision awareness system logged it rest on the driver's own account.

The collision followed directly from this convergence. At 41 m, nothing between the sighting and the impact could have prevented it.

The part dust on the ramp may have played in the truck operator's detection of the light vehicle was examined and could not be resolved. It is discussed under Local Conditions, and nothing from it is carried into this conclusion.

It was extremely likely that both vehicles reached the intersection with neither operator aware of the other, and extremely likely that this contributed to the collision.

The haul truck's front left struck the right side of the light vehicle at 24 km/h

The download recorded the collision at 18:47:12.0 at 24 km/h, 4.6 seconds after the brake application. The truck's accelerometer logged 1.9 g longitudinal.

The scene placed the impact 6.2 m into the ramp from the eastern edge of the spur mouth and 3.1 m out from the inner windrow toe. The damage agreed with this. LV-118 was damaged on the right side from the B pillar forward. HT-407's damage was confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with nothing above the bumper line. Right-side glass and mirror fragments lay from 4.8 m to 14 m down-ramp of the impact position.

The spur joins the ramp from the east, on the inside of a left-hand bend, at an included angle of 71 degrees. A light vehicle emerging from it across the path of a descending truck presents its side, not its front. The impact was therefore a side impact on the light vehicle. The outcome then depended on the light vehicle's structure, the restraint and the impact speed.

The truck operator's braking and steering reduced the impact speed from 37 km/h to 24 km/h. That response is examined under Individual Actions.

It was virtually certain that the truck's front left struck the right side of LV-118 at 24 km/h, and virtually certain that this was the mechanism by which the convergence became an injury.

The light vehicle was pushed 18 m down the ramp and rolled onto its left side

A continuous gouge ran 18.4 m down-ramp from the impact position. At 11.2 m it changed from tyre scuff to body panel gouge, which marks where the vehicle went over. LV-118 came to rest on its left side against the inner windrow, facing down-ramp.

The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout. These are the injuries a side impact and a roll-over produce when the occupant's protection holds. It did hold:

- The roll-over protection structure deformed inward at the right rear mounting but was not breached.
- The driver's seatbelt was found fastened, with loading marks on the webbing.

The same sequence without either of these would have had a different outcome.

The resting side put the driver's door against the ground, so extrication was through the rear of the vehicle. The emergency response rostering that governed the extrication is discussed under Risk Controls. It increased risk but was not found to have contributed to the outcome.

It was virtually certain that LV-118 was pushed down the ramp and rolled onto its left side, and virtually certain that the roll-over was how the collision produced the driver's injuries.

The haul truck's service brakes were degraded and could not achieve the required stopping performance

Degraded service brakes were very unlikely. Brake performance tests on 03/08/2026 and 14/08/2026 gave mean decelerations of 1.9 and 1.92 m/s², the occurrence download gave 1.85 m/s², and no brake, retarder or park brake defect had been raised against HT-407 in the preceding twelve months, so the hypothesis was ruled out.

Conclusions

These findings are made with respect to the occurrence and should not be read as apportioning blame or liability to any organisation or individual.

Findings

Probable cause

- **1.** It was virtually certain that collision awareness alerts were suppressed inside a geofence extended to cover the new intersection, because the S4-LOAD boundary had been moved on 09/06/2026 to a point 180 m up-ramp of RL 312. It was likely that this contributed to the collision. The contribution was not through the alert that did not sound, which would have come 1.8 seconds after braking began. It was through the expectation, built by the absence of alerts on that ramp, that silence meant an empty road. (Safety issue)
- **2.** Both vehicles very likely arrived at the intersection with neither operator aware of the other, and this condition very likely contributed to the collision, since at 41 m of sight line nothing between the convergence and the impact could prevent it.
- **3.** With virtual certainty, the haul truck's front left struck the right side of the light vehicle at 24 km/h, and with virtual certainty that impact was the mechanism by which the convergence became an injury.
- **4.** The light vehicle was, with virtual certainty, pushed 18 m down the ramp and rolled onto its left side, and that roll-over virtually certainly contributed to the driver's injuries, which were limited by a fastened seatbelt and a roll-over protection structure that deformed without being breached.
- **5.** The traffic management guidance, MSG-14, virtually certainly left the review trigger undefined and the interaction zones to the site. The mine was not in breach of the guideline. On balance, and less firmly than the other contributing factors, this discretion likely contributed, because nothing outside the mine required the eleven unreferred roads to be considered by the plan's owner. (Safety issue)
- **6.** Under a works order that recorded management of change as not required, the access spur was, with virtual certainty, cut and opened to traffic with no management of change raised. This very likely contributed by removing the referrals that would have brought the new intersection to the traffic management plan owner, the in-cab road network and the collision awareness geofence configuration. (Safety issue)
- **7.** The spur virtually certainly met the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m and a stopping distance of 96 m. This geometry very likely contributed, because a truck at the posted speed could not stop within the distance available.
- **8.** BIN-TMP-004 virtually certainly required positive radio calls only at listed intersections, and the list was rewritten once a year, with no provision for adding an intersection between reviews. This very likely contributed, since the RL 312 intersection could not have been listed before 12/11/2026. (Safety issue)
- **9.** Neither operator made a positive call on the pit channel before entering the intersection, which was virtually certain and consistent with both the plan and local practice. It was likely that this contributed, a call giving each vehicle knowledge of the other but not separation over 41 m of sight line.

- **10.** It was very likely that the contractor induction was accepted as equivalent, with the site supplement and pit driving assessment never verified, because the access card system had no field for either. This likely contributed. It was discounted because the give-way rule the assessment would have taught was one the driver was already applying, and because the sight line defeated attention on its own. (Safety issue)
- **11.** Without a sign, stop line or give-way control at the mouth, the light vehicle entered the ramp from the spur without stopping and without establishing the truck's position, which was virtually certain. This very likely contributed, although a full stop would have given the driver only a few more metres of sight line, not the 96 m the truck needed.

Contributing factors

The two factors below increased risk. They did not contribute to this occurrence.

- **12.** One emergency response paramedic was, with virtual certainty, rostered on night shift, with the second responder called from the village 11 km from the gatehouse. It was very likely that this arrangement increased risk, as it delayed extrication by approximately 9 minutes and does not change with the severity of the casualty. The delay did not affect this driver's clinical outcome. (Safety issue)
- **13.** The haul truck operator was, with virtual certainty, on night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low. The roster likely increased risk as a standing exposure. The operator's recorded 1.4 second response showed no degradation of his performance on this occasion.

Other findings

- **14.** The hypothesis that the ramp surface was wet and the truck could not stop in the distance available was examined and ruled out. The ramp was dry and the deceleration achieved matched the truck's dry figure.
- **15.** The hypothesis that the haul truck's service brakes were degraded and could not achieve the required stopping performance was examined and ruled out. Brake tests before and after the occurrence and the occurrence download agreed.
- **16.** The truck operator braked and steered into the windrow within 1.4 seconds of sighting. This reduced risk by lowering the impact speed from 37 km/h to 24 km/h and by keeping the light vehicle against the inner windrow rather than over the edge.

Whether dust on the ramp reduced the truck operator's ability to see the light vehicle could not be determined, and no finding is made.

- **Evidence that dust was present:** A water cart pass was 7 minutes overdue. PM10 at the crest monitor was 142 micrograms per cubic metre against a shift mean of 88. The operator described dust hanging on the ramp.
- **Evidence that it did not affect detection:** The light vehicle driver was firm that the air was clear. The operator declined to say the dust made a difference. The reading was below the site trigger of 250 and was taken 180 m away. The sighting at 41 m matched what the geometry gives in clear air.
- **Why it could not be settled:** The evidence was evenly balanced. Nothing at Bindarra measures visibility, and resolving the question would require a measurement of what could be seen from either cab.

Probable cause

The probable cause of this occurrence was very likely the following sequence:

1. The RL 312 access spur was cut and opened to traffic under a works order with no management of change raised.
2. This left an intersection 34 m beyond a crest, with 41 m of sight line against a plan minimum of 120 m. The intersection had no warning sign or give-way control and was absent from the positive radio call list, which was rewritten only once a year.
3. The light vehicle entered the ramp without stopping or establishing the truck's position.
4. Both vehicles arrived at the intersection with neither operator aware of the other.
5. The haul truck struck the right side of the light vehicle at 24 km/h and rolled it onto its left side, seriously injuring its driver.

Contributing factors

- Collision awareness alerts were suppressed inside a geofence extended to cover the new intersection. This likely contributed through the operator's expectation that silence meant an empty road, rather than through the alert itself.
- MSG-14 left the review trigger undefined and the interaction zones to the site. This likely contributed, without any breach by the mine.
- Neither operator made a positive call on the pit channel before entering the intersection, which likely contributed.
- The contractor induction was accepted as equivalent, with the site supplement and pit driving assessment never verified. This likely contributed.

Recommendations

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

Mines Safety Guideline MSG-14 leaves three terms for each site to define. It requires a traffic management plan to be reviewed after any significant change to the road network, but does not say what counts as one. It requires the operator to nominate the intersections and interaction zones where positive communication is required, with no minimum and no basis for choosing them. It allows a light vehicle to be made detectable to a heavy vehicle operator by any one of five means. The plan is a site document. It is not approved by the Inspectorate and is not submitted to it unless requested after an occurrence.

Bindarra's plan met the guideline on the day of the occurrence, and the mine was not in breach. The issue is how wide the discretion is, not any failure by the Inspectorate. At Bindarra, all three terms were settled in one document, reviewed once a year, with no outside review. Eleven of the fourteen new roads cut in Stage 4 since the last review were never referred to the plan's owner.

The guidance likely contributed to two outcomes:

- the RL 312 spur being opened without referral to the plan's owner
- positive calls being required only at intersections listed once a year

This finding is less firm than the investigation's other contributing findings. Against it, nothing in the guideline prevented an out-of-cycle review or a full listing of intersections. The failure to refer the new roads arose inside the mine.

No safety action has been recorded against this issue, and no recommendation is made.

Access spur cut and opened to traffic under a works order, with no management of change raised

It was virtually certain that the access spur to bore DW-14 at RL 312 was cut and opened to traffic under works order WO-26-4471 with no management of change record raised.

CRL-PRO-022 clause 5.2 requires a management of change record before any change to the mine's road network. Clause 5.3 excludes routine earthworks under a works order within an approved pit design where no new plant is introduced. The works order classified the 240 m spur as routine earthworks within the approved Stage 4 design and recorded management of change as not required.

As a result, none of the three referrals required at clause 5.4 was made:

- to the traffic management plan owner
- to the survey department, for the fleet management system road network
- to the contractor's technical services group, for the collision awareness system geofences

This very likely contributed to the occurrence. The new intersection was left out of the plan, the in-cab road network and the geofence configuration.

The issue extended beyond this spur. Fourteen new roads have been cut in Stage 4 since the last plan review. The three raised under a management of change record were referred to the plan's owner. None of the eleven raised under a works order was. The exclusion at clause 5.3 lets a new intersection enter an operating pit without any of the referrals that would otherwise follow it.

No safety action has been recorded against this issue, and no recommendation is made.

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

The mine's contractor management standard, CRL-STD-009, allows a contractor's own general induction to be accepted as equivalent where the contractor holds a current recognition agreement. Merrivale Survey & Geotechnical has held one since 2023. The standard also states that equivalence does not extend to the Bindarra site-specific supplement or to the Bindarra light vehicle pit driving assessment. Every person who will drive in an operating pit must complete both at site, and they may not be substituted. The engaging department must verify both before an access card is issued.

In practice, the gatehouse issued cards against the recognition agreement alone. It held no record of either item and was not asked for one. The access card system did not distinguish a person permitted to drive in an operating pit from one who was not. The standard itself was sound. The system had nowhere for the verification it required to take place.

The light vehicle driver's training file recorded an online module and a 20 minute gatehouse briefing, and neither the supplement nor the pit driving assessment. The shortfall in induction and verification was extremely likely to have existed, and it likely contributed to the light vehicle entering the ramp from the spur without stopping and without establishing the truck's position.

This matters because the pit driving assessment is where a driver learns how the pit's traffic rules work in practice, including positive calls and what to do at a blind intersection. A person could therefore drive on an active haul road without it, and no part of the access system would detect the gap.

No safety action has been recorded against this safety issue.

Positive radio calls required only at listed intersections, and the list is rewritten once a year

The operator's traffic management plan, BIN-TMP-004, required a positive call on the pit channel before a vehicle entered any intersection listed at Appendix B. It required no call anywhere else. An intersection could reach that list only by appearing on the road layout drawing at Schedule 1. Both the drawing and the appendix were reissued only at the annual review. Nothing in the plan allowed an intersection to be added between reviews, and nothing required a new road to be referred to the plan's owner when it was cut.

The RL 312 spur was on neither document. Eleven of the fourteen new Stage 4 roads cut since the last review had never been referred. No out-of-cycle review had been held in the plan's eleven-year history.

The positive call was the mine's own defence against this kind of interaction. At this intersection, the plan's own terms left it without effect. The earliest date on which the spur could have been listed was 12/11/2026, three months after the collision. Had a call been required, it very likely would have influenced the outcome: the pit channel was open, both vehicles were on it, and the dispatcher was tracking the truck's position. The same listing route applies to every road cut between reviews, so the issue is not confined to this intersection.

No safety action has been recorded against this safety issue, and no recommendation has been recorded.

Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

The collision awareness system on the Stage 4 haul fleet gives operator advisories only. Its S4-LOAD suppression geofence was drawn around the loading face on 14/03/2025 to stop nuisance alerts. On 09/06/2026 it was extended northward to a boundary 180 m up-ramp of the RL 312 intersection, which placed the intersection inside the suppressed area.

The contractor's technical services group made the extension as a system configuration task. It was not recorded in the traffic management plan or reviewed against it, and nobody outside that group was told. No person on site had been trained on the geofences. The system's standing report covered alert counts and system health but not suppressions. As a result, nine earlier suppressed events involving a light vehicle tag on the main ramp below the boundary were reported to nobody.

A suppression geofence around a loading face is not itself the problem. The safety issue is that the area where the system stays silent could be changed without reference to the traffic management plan and without the change being made known to those relying on the system.

It is likely that this contributed to the collision. On this occasion the suppressed alert would have come after the operator had begun braking. Its influence lay instead in the expectation it had built: long silence on that ramp taught operators that silence meant an empty road, and the operator carried that expectation into the crest. The same condition applies wherever the fleet operates inside a suppressed area.

No safety action has been recorded against this safety issue.

One emergency response paramedic rostered on night shift, with the second responder called from the village

The Bindarra night shift was rostered with one qualified mines rescue paramedic, based at the surface medical facility. The second responder, who had to be present before any extrication began, was called in from the accommodation village 11 km from the gatehouse. It is virtually certain that this arrangement was in place on the night of the occurrence. The roster, the emergency response log and the post-incident review agreed on the times.

Dispatch paged the team at 02:48. The first response vehicle reached the scene at 02:54. The second responder arrived with the hydraulic rescue tool at 03:07. The driver was extricated at 03:14, 27 minutes after the collision. The post-incident review found that the extrication was delayed by approximately 9 minutes while the team waited for the second responder and the tool. It also found that the delay did not affect the clinical outcome, because the injuries were stable and the driver was conscious throughout.

The collision, the injuries and their severity were all complete before the response was paged. It is unlikely that the response arrangement had any bearing on this occurrence or its outcome.

It is very likely that the arrangement is nonetheless a safety issue. It is a structural gap in a last line of defence, present on every night shift. The site's extrication capability at night is one person plus a drive from the village, and nothing in the arrangement changes with the severity of the casualty. A fuel fire, an unconscious casualty, or an airway that needed the vehicle opened within the first ten minutes would make the same wait far more consequential.

No safety action has been recorded against this issue.

Distribution

Title	Name
General Manager, Bindarra Gold Mine	T. Brandner
Registered Manager (statutory)	S. Kirkhope

Mining Manager and owner of BIN-TMP-004	H. Quaile
Manager Health and Safety, Corella Resources	L. Farquhar
Project Manager, Warrigal Mining Services	D. Mokoena
Superintendent Dewatering, Corella Resources	P. Nyaguse